

Discussion of Broxterman, Larson, Yezer “Characteristics of a Sufficient Statistic to Measure City Housing Prices”

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Paper Contributions

- ▶ Shows heterogeneous price growth across space w/in metros
 - ▶ Builds on e.g. Landvoigt, Piazzesi, + Schneider
- ▶ Offers a new methodology for aggregating to the metro level
- ▶ Discussion of aggregating across units vs hedonic quality
 - ▶ Salient to the “shoebox condo” debate

Target for Laspeyre's index: units

$$\frac{\Delta r_t^*}{r_{t-1}} = \frac{\sum_k r_t(k) h_{t-1}(k) / H_{t-1}(k)}{\sum_k r_{t-1}(k) h_{t-1}(k) / H_{t-1}(k)}.$$

Aggregating across tracts, authors propose:

$$\Delta P_{jt} = \sum_i W_{ijt} \Delta P_{it}$$

This works if $W_{ijt} = \sum_{k \in i} \frac{h_{t-1}(k)}{H_{t-1}(k)}$ for all i .

- ▶ What units are h in? Rooms? Value? Units?
- ▶ Warrants further discussion
 - ▶ Not easy to see use where luxury homes get huge weight
 - ▶ Tiny condos getting much cheaper in YYZ helpful?
- ▶ Also why are transaction weights non-representative?
 - ▶ Table 2 shows weights matter
 - ▶ New homes a good example (excluded!)
 - ▶ But what data generating process causes this?

Emphasis on location as heterogeneity

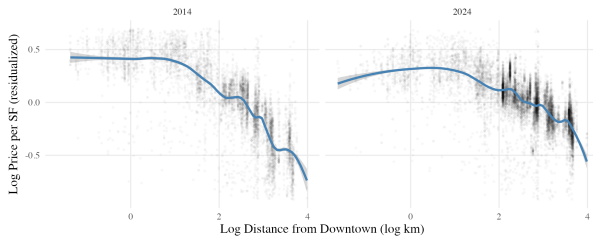
- ▶ Result from familiar monocentric city model
 - ▶ $\frac{dr(k)}{dk} = -\frac{\frac{d\text{commute cost}}{dk}}{h_t}$
 - ▶ If I move a bit further, $|h\dot{r}| = |\text{commute cost}|$
- ▶ Also if constant (construction + option premium) at boundary
 - ▶ $d[r(k) - r(b(t))] = \theta \dot{b} = dr(k) = \text{constant}$
 - ▶ Then need constant value in distance
 - ▶ Some condition on income and price elasticities
- ▶ But linear commute cost ??
 - ▶ 41st to 61st St in Manhattan vs random mile in the suburbs
- ▶ Suppose $\ln p(h, k) = \ln h + \theta \log \frac{b}{k}$, then $g'(k) = 0$
(with light conditions.)

My empirical work plan for discussion

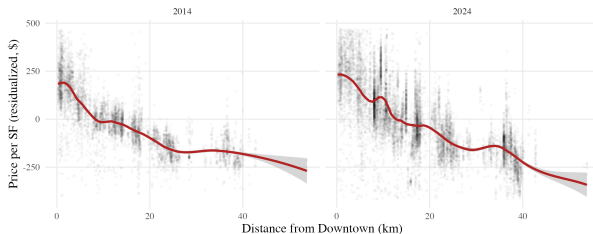
- ▶ Show clearly log beats linear price gradient in Vancouver
- ▶ Explain that location not the logical heterogeneity to focus on
 - ▶ Or at least that this is overblown “Unless highly restrictive preference assumptions are imposed, appreciation rates cannot be spatially invariant.”
- ▶ But not exactly what happened ...

Vancouver condo prices by year: distance gradients

A. Log-Log Specification



B. Linear-Linear Specification



Vancouver Condo: fit

- ▶ Log-log fits better than level-level 2014
- ▶ Subsequent years
 - ▶ Gradient flattens a lot
 - ▶ Action is at outer edge
 - ▶ Levels gets to be much better fit
- ▶ So $g'(k) \neq 0$ as authors emphasize
- ▶ Reg level P on log distance generally fits by far best
 - ▶ inverse of old literature
 - ▶ but doesn't capture flattening
 - ▶ True for single and condo
- ▶ Box-Cox results for λ :
 - ▶ Appx. .6 (2!) with level (log) on RHS (similar condo/SF)

Pairwise correlations among Vancouver vs Suburban types

Type matters more than location?

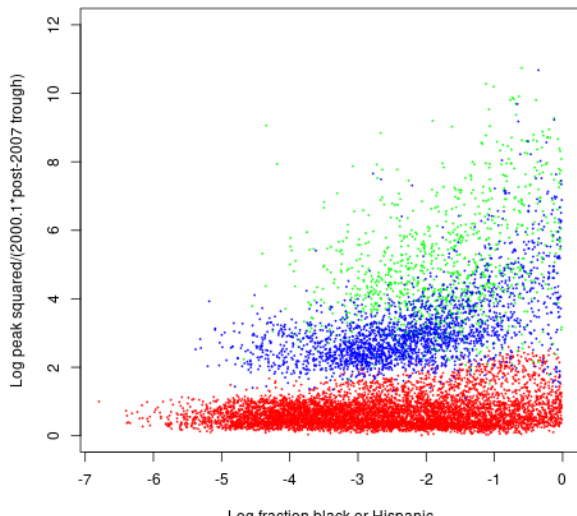
	VanSingle	VanCondo	FVSingle	FVCondo
VanSingle	1.00	0.71	0.82	0.54
VanCondo	0.71	1.00	0.71	0.70
FVSingle	0.82	0.71	1.00	0.71
FVCondo	0.54	0.70	0.71	1.00

- ▶ Log index growth 2014-2024: FV condo: 10.7%
- ▶ GV single: 5.9%
- ▶ Story: “flight from quality” after foreign buyer ban?

GFC: heterogeneity driven by subprime propensity

Zip code price movements boom bust (source, Zillow)

Red: Flyover, Blue: Coastal, green: Sand State



Transaction weight: spatial smoothing vs tract indexes?

- ▶ Index = sum over tracts tract weight times tract price change
- ▶ Start from equal tract weights
- ▶ Then transaction weight = $1/\text{number of transactions in tract}$
 - ▶ This seems like an invitation to outlier problems
- ▶ Suggestion: borrow from kernel estimation literature
- ▶ This can extend to multiple dimensions with cell weights
 - ▶ Price point (target income/credit constraints)
 - ▶ Investor/owner occupier (condo/single)
 - ▶ ...
- ▶ Optimal borrowing from neighbors?
 - ▶ Probably more precision
 - ▶ Bad for certain purposes (e.g. subunit correlations)

One motivation for messed up weights: market hotness

- ▶ Winners sell, losers held
- ▶ Does this bias up growth? Down volatility
 - ▶ Seems like a worthy exercise for paper
- ▶ Examples:
 - ▶ West Van detached pre/post foreign buyer stuff
 - ▶ GFC boom/bust minority neighborhoods

What is the question indexes answer?

- ▶ What decision-maker needs to aggregate to 1-type-metro-level?
 - ▶ Mark-to-market LTV distribution? X (nonlinearity)
 - ▶ Real estate pro? X (comp set ...)
 - ▶ Portfolio efficiency for investor? X
 - ▶ Workout decision for lenders? X
 - ▶ Government transfer tax forecast? But tx weights!
 - ▶ CPI (incumbent vs turnover?)
- ▶ Has anyone messed up an optimization with wrong index info?

AHS as source for index comparisons?

- ▶ 2015-2019 Table 1 shows Atlanta 18% vs 13% per year appreciation
- ▶ FHFA per FRED says about 7%
- ▶ Washington, D.C. Table 1 says about 2%
- ▶ FHFA per FRED says about 3.7%.
- ▶ Aggregate Zillow zip indexes by metro central/suburbs?
- ▶ Strong suggestion: use better data and see if differences as large

Summary

- ▶ Important takeaways from paper
 - ▶ Price growth not constant across submarkets
 - ▶ Spatial differences in growth stand out
 - ▶ Aggregation weights depend on target unit of measure
 - ▶ Transaction indicator unlikely optimal weighting
- ▶ Suggestions
 - ▶ Consider whether spatial differentiation warrants focus
 - ▶ Sane models *can* imply constant appreciation in distance
 - ▶ My evidence mixed and weird on gradient
 - ▶ Discuss this literature (Mills, McMillen, Coulson, ...) more
 - ▶ Use better data and work with kernel smoothing techniques
 - ▶ Inverse transaction weights likely problematic
 - ▶ Do more work to convince that large aggregates have value